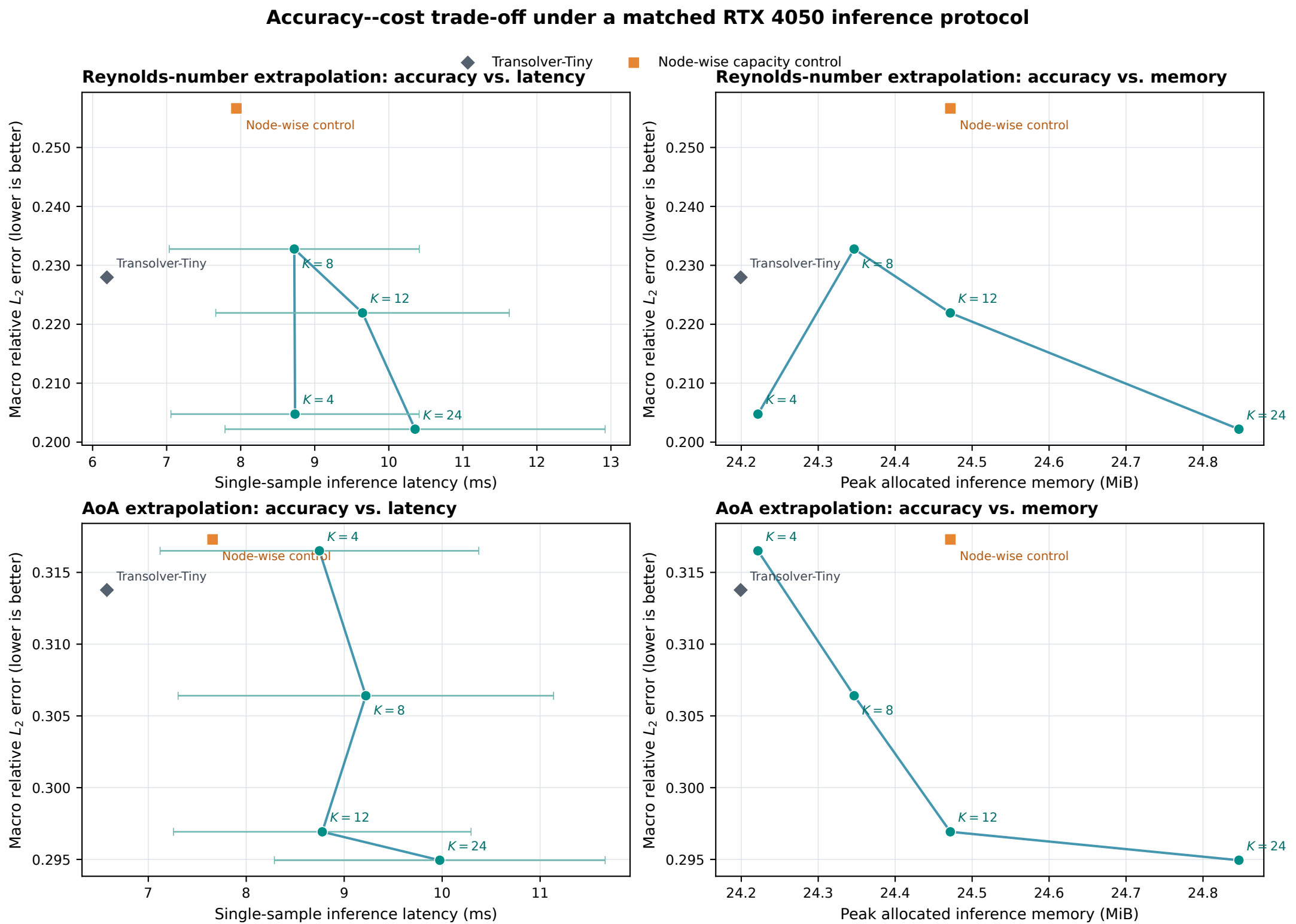
Accuracy--cost trade-off under a matched RTX 4050 inference protocol
Transolver-Tiny
Node-wise capacity control
Reynolds-number extrapolation: accuracy vs. latency
Reynolds-number extrapolation: accuracy vs. memory
AoA extrapolation: accuracy vs. latency
AoA extrapolation: accuracy vs. memory
Node-wise control
Transolver-Tiny
K = 4
K = 8
K = 12
K = 24
Macro relative L_2 error (lower is better)
Single-sample inference latency (ms)
Peak allocated inference memory (MiB)
Fixed $N = 4096$, $S = 64$, $d = 64$, $L = 4$; mixed precision; 20 warm-up and 100 measured forwards. KNN indices are precomputed and construction time is excluded.